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1 Goal and Area of Application

The use of a dose-response matrix allows you to identify and quantify additive, synergistic and antagonistic dose regions in the complete interaction landscape.

The interaction landscape offers an increased power to differentiate between various classes of drug combinations, and may therefore provide an improved means for understanding their mechanisms of action toward clinical translation.This SOP describes procedures to perform a drugs combination screening assay for cells seeded in 384-well plates in a dose-response matrix experiment.

2 Definitions and Abbreviations

SOP	Standard Operating Procedure
CP	Cell Plate
MP	Master Plate
EP	Experimental Plate
RT	Room Temperature
DMSO	Dimetthyldulfoxide
FBS	Fetal Bovine Serum
PBS	Phosphate-buffered saline
PT0	plate measured at time 0
Lum	luminescence

3 Devises, Materials and reagents

3.1 Devises

Equivalent devices can be used, but should be checked to ensure operational condition and suitability for intended use. Deviations are at own risk and not advised by the authors of this SOP.

Device name	Supplier
ML STAR liquid handling	Hamilton
BIOTEK Gen5	Agilent tech
Cell incubator	General

TC20™ Automated Cell Counter	Biorad
Sterile Hood	General

3.2 Materials

Equivalent materials can be used, but should be checked to ensure operational condition and suitability for intended use.

Product	Supplier	Cat. No
Tips	384 50µl CO-RE TipTips with filters-Hamilton	235844
	1000 µL CO-RE® Tip. Tips with filters-Hamilton	235905
Tubes	5mL-Aptaca SpA	10470/SG
Reagent Container	60 mlSelf-standing, with lid-Hamilton	56694-01
Reservoir	300 mL-Axygen	RES-SW384-FX-SI
Assay plate	VIEWPLATE-384 TC, White, clear bottom, sterile TC treated-Revvity	6007480
Drug plate Storage	Storage Microplates Axygen	Axygen™ P384240SQCS

3.3 Reagents

Equivalent reagents can be used, but should be checked to ensure operational condition and suitability for intended use. Due to their dependency on the specific cancer cell line employed in the SOP, certain materials will not be exhaustively delineated in Supplier and Cat.No.; Nonetheless, initiating experimentation with a well-characterized cell line is crucial to guaranteeing optimal growth conditions.

Product	Supplier	Cat. No
Medium (e.g. DMEM)	general	-
Phosphate-buffered saline (PBS, 1X), sterile-filtered	general	-

Trypsin-EDTA (0.05%), phenol red	general	-
Medium supplements (e.g. PEN 10.000U/ml-Strept10.000µg/ml and FBS)	general	-
Positive Control	Cytotoxic compounds e.g.: Doxorubicin; LBH	-
DMSO	general	-
Assay	CellTiter-Glo® Luminescent Cell Viability Assay Promega	G7572
	CellTiter-Glo® 3D Cell Viability Assay Promega	G9683

4 Preliminary Procedures

Before starting the main procedures, it will be necessary to define, through preliminary experiments, the number of cells and the dosage curve of the drugs tested in combination that must be used in the subsequent experimental procedure.

4.1 Cell number definition

Cells are harvested from 75cm^2 flask at 80 % confluency by washing them once using 5mL RT PBS (1X), afterwards incubating the cells with 2mL Trypsin-EDTA for 5 min at 37°C, collecting them in 15 mL falcon by adding medium to neutralize Trypsin activity and centrifuging (5min x 1200rpm).

The cells pellet is suspended in 5 mL of prewarmed cell culture media and counted using TC20™ Automated Cell Counter.

Cells are diluted to allow the seeding of cells in 5 different concentration (usually 1200-1000-800-600-400 cells/well) in 32 µL for each well. Each concentration is plated in sixfold on a cell carrier 384-plate (Revvity).

Cells are incubated for 96 h at 37°C and 5% CO^2. After 96h (the timing of drug combination screening) of incubation, the number of viable cells plated is analyzed by adding 32 µL of CellTiter-Glo® Promega reagent (previously diluted with a 1:2 ratio of milliQ water) to each well and mix for 2 minutes (in the dark) on an orbital shaker to induce cell lysis (not included in liquid handling station, manually). The plate is incubated for 10 min in the dark again at RT and luminescence signaling is measured using the Cytation5 micro plate reader (BioTek). The parameters for bioluminescence reding are a GAIN of 135 and an integration time of 0.25–1 second per well, as recommended by the protocol.
The best cell dosage to select should be higher than 10000 and lower than 1000000.

4.2 Drug dosage curve definition

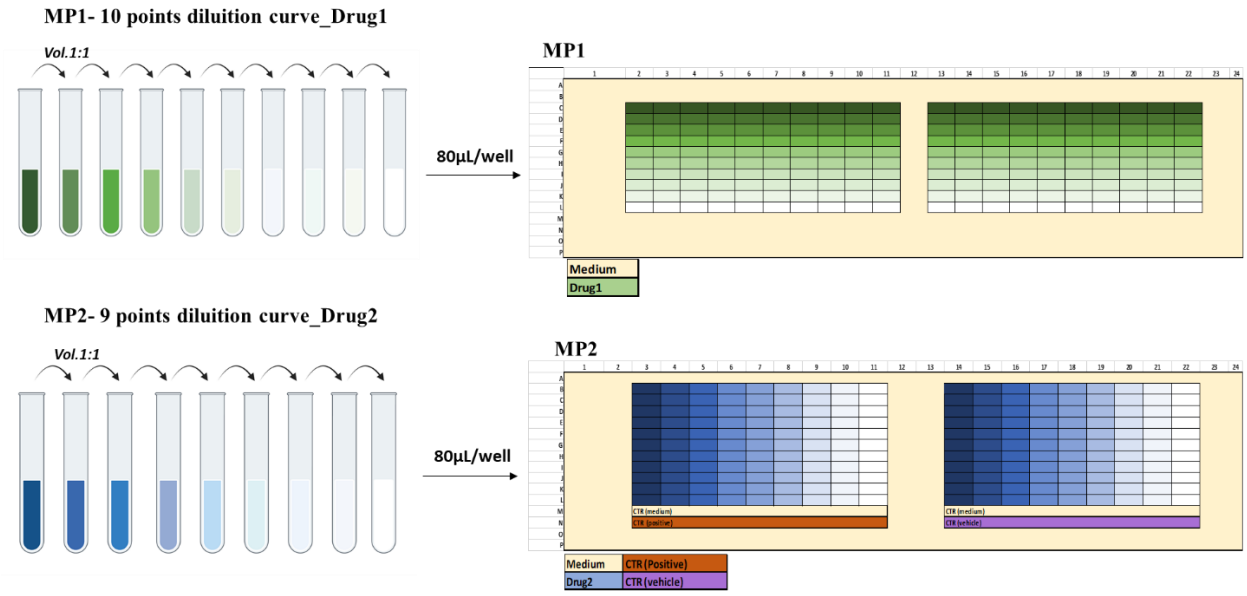
This drug combination SOP must be conducted upon dose-response experiments of individual drugs (referr to SOP for *IC50 identification*). It will be necessary to establish an efficacy curve for the drugs under examination. The experimental curve for each individual drug under examination will encompass an antiproliferative effect ranging from 20% to 80% when administered individually.

5 Procedure

5.1 Master Plate generation

This procedure is designed to combine dilution series of two drugs in a matrix, mixed to measure their anti-proliferative effect when admistred in combination. We generated two master plates (MP1 and MP2) containing the drug dilution (definited before) in monoplicate, to be added to cell plate (CP) in duplicate.
The dilution series of the two drugs will be added to MP1 and MP2. In detail, each point curve will be transferred in MP at a concentration 10 times higher (10X) than the final one to be reached on the cells. The first tube contains the drug at dedicated concentration in a volume that is twice compared to the others, since a serial dilution is performed to obtain 10 points

for Drug1 and 9 points for Drug2. Subsequently from each tube 80 µL are dispensed in 384-MP in monoplicate, as schematized in the example below:



The second MP contains also the negative and positive controls, respectively vehicle (i.e DMSO) and a drug with a known cytotoxic outcome (i.e. Doxorubicin) which approximate 100% of cell proliferation inhibition.

5.2 Cell seeding

Cells are harvested from 75cm^2 flask at 80 % confluency by washing them once using 5mL RT PBS (1X), afterwards incubating the cells with 2mL Trypsin-EDTA for 5 min at 37°C, collecting them in 15 mL falcon by adding medium to neutralize Trypsin activity and centrifuging (5min x 1200rpm). The cells pellet is suspended in 5 mL of prewarmed cell culture media and counted using TC20™ Automated Cell Counter (Biorad).

Cells are diluted to a definited concentration per mL (i.e.800 cells/well) (as previously determined in section 4.1) and 32 µL of this suspension is added to each well cell carrier 384-plate (Revvity). The experimental plate (EP) number is variable depending on the

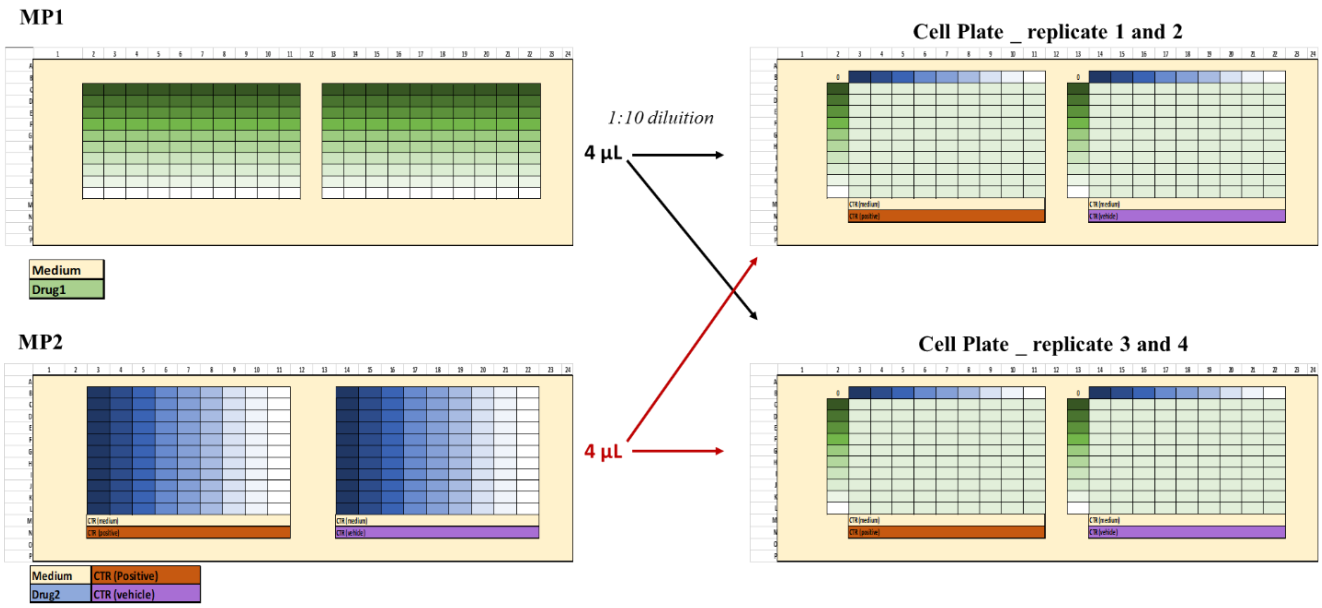
experimental set-up. Just remember to add a plate to measure the untreated T0 (PT0). Cells are incubated for 24 h at 37°C and 5% CO_2 .

5.3 Cells status check post-seeding and before the tratemnt

Before proceeding with the treatment, it will be necessary to assess the proliferative status of the cells. This procedure is essential for subsequent growth rate calculations.
At the time of treatment (24h after cells seeding), defined T0, a plate for each tested cell line will be assayed by adding 32 µL of CellTiter-Glo® reagent (previously diluted 1:1 in milliQ water) to each well and mix for 2 minutes (in the dark) on an orbital shaker to induce cell lysis (not included in liquid handling station, manually). The plate is incubated for 10 min in the dark again at RT and luminescence signaling is measured using the Cytation5 micro plate reader (BioTek). The parameters for bioluminescence reding are a GAIN of 135 and an integration time of 0.25–1 second per well, as recommended by the protocol.

5.4 Cell Plates treatment

The drugs from each MP will be added to EP and repeated for n cell plate (EP). In details, a previously determined number of cells/well will be seeded in 32µl of medium and 24h later 4µL from every well of each MP will be added to 2 identical CP in order to obtain quadruplicates, reaching a final volume of 40µL in each well at the end of treatment, as schematized below:



5.5 Read out Measurements and data analysis

The CellTiter-Glo® Luminescent Cell Viability Assay (Promega) is a homogeneous method of determining the number of viable cells in culture based on quantitation of the ATP present, an indicator of metabolically active cells. The CellTiter-Glo® Assay is designed for use with multiwell formats, making it ideal for automated high-throughput screening (HTS), cell proliferation and cytotoxicity assays. The homogeneous assay procedure involves adding the single reagent (CellTiter-Glo® Reagent) directly to cells cultured in a serum-supplemented medium. Cell washing, removal of medium and multiple pipetting steps are not required. The system detects as few as 15 cells/well in a 384-well format in 10 minutes after adding reagent and mixing.

The luminescence (lum) is recorded by Gen5 Software. The parameters for bioluminescence reding are a GAIN of 135 and an integration time of 0.25–1 second per well, as recommended by the protocol. Finally, the measures from T0 Plate and the experimental plates are exported in excel file for further evaluation.

5.6 Quality Control parameters

Z-factor (z-F) , a screening window coefficient, a measure of statistical effect size and calculated as the means (μ) and standard deviation (σ) of both positive (p) and negatives (n) controls:

$$z\text{-}F = 1-3(\sigma_p+\sigma_n)/|\mu_p-\mu_n|$$

The coefficient of assay signal dynamic range and the data variation is associated with the signal measurements, and therefore suitable for assay quality assessment. The Z-factor is a dimensionless, simple statistical characteristic for each HTS assay. The Z-factor used in assay optimization and validation: provides comparison and evaluation of the quality of assays (Zhang JH, J. Biomol. Screen 1999)

Acceptable Z-factor value ranges between 0.5 and 1. The plate is invalidated if Z-factor value calculated in below 0.3. The Z-factor value can never exceed 1 (*Zhang JH, et al. J Biomol Screen. 1999;4(2):67-73*).

6 Data Analisys

The synergistic antitumor effect will be analyzed by SynergyFinder (refer to Synergy finder data analysis SOP).

7 Further applicable documents

SOP for *IC50 identification*

SOP for Synergy finder data analisys.